

Analysis First-Year Exam, August 2017, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $f : [0, 1] \rightarrow [0, 1)$ be continuous. Decide whether it follows that

$$\int_0^1 \frac{1}{1 - f(t)} dt = \sum_{n=0}^{\infty} \int_0^1 f(t)^n dt,$$

giving proof or counterexample as appropriate. [Note the intervals carefully.]

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be nonconstant; for each positive integer n and each real t define $f_n(t) = f(nt)$. Prove that there exists no $\varepsilon > 0$ such that $(f_n)_{n=1}^{\infty}$ is equicontinuous on the interval $(-\varepsilon, \varepsilon)$.
3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Either
- (i) show that there exists a sequence of polynomials converging pointwise to f on $\mathbb{R}$
 - (ii) or show that there need not exist a sequence of polynomials converging uniformly to f on $\mathbb{R}$.
4. Let $(f_n)_{n=1}^{\infty}$ be a sequence of measurable real-valued functions. Prove that each of the following sets is measurable:
- (i) $A = \{\omega : f_n(\omega) \rightarrow \infty \text{ as } n \rightarrow \infty\}$;
 - (ii) $B = \{\omega : f_n(\omega) \text{ is eventually irrational}\}$;
 - (iii) $C = \{\omega : f_n(\omega) > 0 \text{ for infinitely many } n\}$.
5. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be locally Lebesgue integrable and assume that $f(t) \rightarrow 1$ as $t \rightarrow \infty$. Prove that for each positive integer n we may define

$$a_n = \frac{1}{n} \int_0^{\infty} e^{-t/n} f(t) dt \in \mathbb{R}$$

and prove that $a_n \rightarrow 1$ as $n \rightarrow \infty$.

Suggestion: Say $|f(t) - 1| \leq \varepsilon$ whenever $t \geq A$ and split the integral.